

Machine Learning with Big DATA Assignment 1

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GIT Repo Link:

https://github.com/aditya85599/MLBD_assign1

1. Run and show the working of the WordCount [example](#) as shown on the Apache Hadoop website as well as discussed in class earlier. (No need to send code for this. Just output to show that it is working is enough).

O/p :

```
(base) admin@Aditya-2 logs % hdfs dfs -cat /output/part-r-00000
2026-02-13 16:23:55,642 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
BIG 1
Hello 1
data 1
first 1
hadoop 2
is 1
my 1
this 1
world 1
(base) admin@Aditya-2 logs %
```

2. What will the output pairs look like?

O/p

Lines	Key --> Text	Value --> IntWritable
We're up all night till the sun	We're	1
	up	1
	all	1
	night	1
	till	1
	the	1
	sun	1

What will be the types of keys and values of the input and output pairs in the Map phase?

Input Key Type	LongWritable
Input Value Type	Text
Output Key Type	Text
Output Value Type	IntWritable

3. For the Reduce phase, some of the output pairs will be the following:

("up", 4)
("to", 3)
("get", 2)
("lucky", 1)

What will the input pairs look like?

Hadoop automatically perform shuffle and sort, it group all the values with same key.

("Up", 1)
("Up", 1)
("Up", 1)
("Up", 1)

Into ("Up", [1,1,1,1]) now reducer will sum all the values and give O/p ("Up", 4)

similarly ("to", 3), ("get", 2) and ("lucky", 1)

What will be the types of keys and values of the input and output pairs in the Reduce phase?

Stage	Key Type	Value Type
Reduce Input	Text	Iterable<IntWritable>
Reduce Output	Text	IntWritable

4. [Mapper](#) maps input key/value pairs to a set of intermediate key/value pairs.

From ques 2 -->

```
public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException
```

From Ques 3 -->

```
public void reduce(Text key, Iterable<IntWritable> values,
    Context context)
    throws IOException, InterruptedException
```

Arguments to pass

- job.setOutputKeyClass(**Text.class**);
- job.setOutputValueClass(**IntWritable.class**)

5. Write the map() function. We want to make sure to disregard punctuation: to this end, you can use String.replaceAll(). In order to split lines into words, you can use a StringTokenizer.

```
public static class Map
    extends Mapper<LongWritable, Text, Text, IntWritable> {

    private final static IntWritable one = new IntWritable(1);
```

```

private Text word = new Text();

public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {

    String line = value.toString().toLowerCase();
    line = line.replaceAll("[^a-zA-Z0-9\\s]", "");

    StringTokenizer tokenizer = new StringTokenizer(line);

    while (tokenizer.hasMoreTokens()) {
        word.set(tokenizer.nextToken());
        context.write(word, one);
    }
}
}

```

6. Write the reduce() function. When you're done, make sure that compiling the project doesn't produce any errors.

```

public static class Reduce
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }

        context.write(key, new IntWritable(sum));
    }
}

```

7. `hdfs dfs -getmerge` command, `hdfs dfs -ls /DM184MB_Output`

```
(base) admin@Aditya-2 MLBD_assign_1 % hdfs dfs -getmerge /DM184MB_Output DM184MB_output.txt

2026-02-14 16:08:24,088 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
(base) admin@Aditya-2 MLBD_assign_1 % head -20 DM184MB_output.txt

0      44
00      9
000      2
00000195      1
00000222      1
00000231      1
0000025 1
0000611 1
0001118 1
00013 1
```

hdfs dfs -ls /DM184MB

```
(base) admin@Aditya-2 MLBD_assign_1 % hdfs dfs -ls /

2026-02-14 16:06:22,682 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 7 items
drwxr-xr-x - admin supergroup 0 2026-02-14 16:03 /DM184MB
drwxr-xr-x - admin supergroup 0 2026-02-14 16:04 /DM184MB_Output
drwxr-xr-x - admin supergroup 0 2026-02-14 15:15 /anotherinput
drwxr-xr-x - admin supergroup 0 2026-02-14 15:17 /anotheroutput
drwxr-xr-x - admin supergroup 0 2026-02-13 16:23 /input
drwxr-xr-x - admin supergroup 0 2026-02-13 16:23 /output
drwxr-xr-x - admin supergroup 0 2026-02-14 15:11 /wcoutput
(base) admin@Aditya-2 MLBD_assign_1 % hdfs dfs -ls /DM184MB_Output
```

Job running in YARN UI (8088)



Nodes of the cluster

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources	Reserved Resources	Physical Mem %
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:8 GB, vCores:8>	<memory:0 B, vCores:0>	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
1	0	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0

Show 20 entries

Node Labels	Rack	Node State	Node Address	Node HTTP Address	Last health-update	Health-report	Containers	Allocation Tags	Mem Used	Mem Avail	Phys Mem Used %	VCores Used	VCores Available
/default-rack		RUNNING	localhost:51348	localhost:8042	Sat Feb 14 16:07:19 +0530 2026		0		0 B	8 GB	0	0	8

Showing 1 to 1 of 1 entries

First P

8. Directories in HDFS do not have a replication factor because replication applies only to file data blocks stored on DataNodes. A directory in HDFS does not contain any data blocks; it only stores metadata in the NameNode. Since there are no physical data blocks associated with a directory, there is nothing to replicate. Therefore, replication factor is applicable only to files and not to directories.

```
(base) admin@Aditya-2 MLBD_assign_1 % hadoop fs -ls /

2026-02-14 16:32:32,749 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 7 items
drwxr-xr-x - admin supergroup 0 2026-02-14 16:03 /DM184MB
drwxr-xr-x - admin supergroup 0 2026-02-14 16:04 /DM184MB_Output
drwxr-xr-x - admin supergroup 0 2026-02-14 15:15 /anotherinput
drwxr-xr-x - admin supergroup 0 2026-02-14 15:17 /anotheroutput
drwxr-xr-x - admin supergroup 0 2026-02-13 16:23 /input
drwxr-xr-x - admin supergroup 0 2026-02-13 16:23 /output
drwxr-xr-x - admin supergroup 0 2026-02-14 15:11 /wcoutput
(base) admin@Aditya-2 MLBD_assign_1 %
```

9.

Before : Default configuration

```
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=8312639
File Output Format Counters
  Bytes Written=856349
Total Execution Time: 3856 ms
(base) admin@Aditya-2 MLBD_assign_1 %
```

After : 1MB configuration

```
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=8341311
File Output Format Counters
  Bytes Written=856349
Total Execution Time: 2798 ms
(base) admin@Aditya-2 MLBD assign_1 %
```

Split Size	Execution Time
Default	3856ms
1MB	2798ms

- Smaller splits increase number of mappers
- Too many small splits increase scheduling overhead
- Optimal split size depends on cluster resources

10. Book Metadata Extraction and Analysis

Task:

1. **Metadata Extraction:** Extract the following metadata from the text column of each book in the books_df DataFrame:

- title
- release_date
- language
- encoding

file_name	title	release_date	language	encoding
hdfs://localhost:9000/D184MB/115.txt	United States Census Figures back to 1630	2008	English	ASCII
hdfs://localhost:9000/D184MB/27.txt	Far from the Madding Crowd	1992	English	ASCII
hdfs://localhost:9000/D184MB/124.txt	Deuterocanonical Books of the Bible	2008	English	ASCII
hdfs://localhost:9000/D184MB/144.txt	The Voyage Out	2006	English	ASCII
hdfs://localhost:9000/D184MB/26.txt	Paradise Lost	1992	English	ASCII
hdfs://localhost:9000/D184MB/257.txt	Troilus and Criseyde	1995	English	ASCII
hdfs://localhost:9000/D184MB/228.txt	The Aeneid	2008	English	ASCII
hdfs://localhost:9000/D184MB/178.txt	Confidence	2006	English	ASCII
hdfs://localhost:9000/D184MB/222.txt	The Moon and Sixpence	2007	English	ASCII
hdfs://localhost:9000/D184MB/223.txt	The Wisdom of Father Brown	1995	English	ASCII
hdfs://localhost:9000/D184MB/306.txt	The Early Short Fiction of Edith Wharton, Part 2 (of 10)	1995	English	ASCII
hdfs://localhost:9000/D184MB/209.txt	The Turn of the Screw	1995	English	ASCII
hdfs://localhost:9000/D184MB/16.txt	Peter Pan	1991	English	ASCII
hdfs://localhost:9000/D184MB/234.txt	Child Christopher	2008	English	ASCII
hdfs://localhost:9000/D184MB/271.txt	Black Beauty	2006	English	ASCII
hdfs://localhost:9000/D184MB/32.txt	Herland	1992	English	ASCII
hdfs://localhost:9000/D184MB/163.txt	Flower Fables	2008	English	ASCII
hdfs://localhost:9000/D184MB/210.txt	An International Episode	2008	English	ASCII
hdfs://localhost:9000/D184MB/291.txt	The Golden Age	2008	English	ASCII
hdfs://localhost:9000/D184MB/11.txt	Alice's Adventures in Wonderland	1994	English	ASCII

2. Analysis:

- Calculate the number of books released each year.
- Find the most common language in the dataset.
- Determine the average length of book titles (in characters).

release_date	count
1991	4
1992	7
1994	10
1995	15
2004	42
2006	2
2007	25
2008	6
2010	90
2011	3
2012	1
2015	2
2015	1

```
26/02/14 17:47:33 INFO SparkContext: Invoking stop() from shutdown hook
26/02/14 17:47:33 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:539.
26/02/14 17:47:33 INFO SparkUI: Stopped Spark web UI at http://192.168.1.24:4041
```

language	count
English	200
Latin	6
	2

```
26/02/14 19:01:18 INFO SparkContext: Invoking stop() from shutdown hook
26/02/14 19:01:18 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:539.
26/02/14 19:01:18 INFO SparkUI: Stopped Spark web UI at http://192.168.1.240:4041
26/02/14 19:01:18 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/14 19:01:18 INFO MemoryStore: MemoryStore cleared
26/02/14 19:01:18 INFO BlockManager: BlockManager stopped
26/02/14 19:01:18 INFO BlockManagerMaster: BlockManagerMaster stopped
26/02/14 19:01:18 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/02/14 19:01:18 INFO SparkContext: Successfully stopped SparkContext (Uptime: 7557 ms)
(venv) (base) admin@Aditya-2 MLBD_assign_1 %
```

```

+-----+
|avg(length(title))|
+-----+
| 23.39903846153846|
+-----+

26/02/14 19:02:22 INFO SparkContext: Invoking stop() from shutdown hook
26/02/14 19:02:22 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:539.
26/02/14 19:02:22 INFO SparkUI: Stopped Spark web UI at http://192.168.1.240:4041
26/02/14 19:02:22 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/14 19:02:22 INFO MemoryStore: MemoryStore cleared
26/02/14 19:02:22 INFO BlockManager: BlockManager stopped
26/02/14 19:02:22 INFO BlockManagerMaster: BlockManagerMaster stopped
26/02/14 19:02:22 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/02/14 19:02:22 INFO SparkContext: Successfully stopped SparkContext (Uptime: 7440 ms)
(venv) (base) admin@Aditya-2 MLBD_assign_1 %

```

Regular expressions were used to extract structured metadata fields such as title, release date, language, and encoding from the Project Gutenberg header. Each regex pattern matched a specific metadata label followed by its value using capturing groups.

However, regex-based extraction has limitations due to inconsistent formatting, case sensitivity, multi-line fields, and missing metadata. Extracted data may contain inconsistencies, null values, or incorrect matches.

In real-world scenarios, these issues would be handled through data cleaning, normalization, case-insensitive matching, validation checks, and possibly using structured metadata sources instead of raw text parsing.

11.

```

26/02/14 19:35:00 INFO CodeGenerator: Code generated in 15.135625 ms
26/02/14 19:35:00 INFO CodeGenerator: Code generated in 2.623125 ms

+-----+-----+-----+
|book1|book2|cosine_similarity|
+-----+-----+-----+
|hdfs://localhost:9000/D184MB/108.txt|hdfs://localhost:9000/D184MB/221.txt|0.9991771501221716|
|hdfs://localhost:9000/D184MB/10.txt|hdfs://localhost:9000/D184MB/30.txt|0.9987882622903789|
|hdfs://localhost:9000/D184MB/117.txt|hdfs://localhost:9000/D184MB/156.txt|0.9909735039505784|
|hdfs://localhost:9000/D184MB/133.txt|hdfs://localhost:9000/D184MB/298.txt|0.9817156238580168|
|hdfs://localhost:9000/D184MB/204.txt|hdfs://localhost:9000/D184MB/223.txt|0.9770610790733313|
+-----+-----+-----+

only showing top 5 rows
26/02/14 19:35:00 INFO SparkContext: SparkContext is stopping with exitCode 0 from stop at NativeMethodAccessorImpl.java:0.
26/02/14 19:35:00 INFO SparkUI: Stopped Spark web UI at http://192.168.1.240:4041
26/02/14 19:35:00 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/14 19:35:00 INFO MemoryStore: MemoryStore cleared
26/02/14 19:35:00 INFO BlockManager: BlockManager stopped
26/02/14 19:35:00 INFO BlockManagerMaster: BlockManagerMaster stopped

```

- Explain TF and IDF. Why is TF-IDF useful for weighting the importance of words in a document?**

TF measures importance inside a document.
IDF reduces importance of globally common words.
TF-IDF gives balanced word weighting.
- How is cosine similarity calculated? Why is it appropriate for comparing documents represented as TF-IDF vectors?**

Cosine similarity measures angular similarity between TF-IDF vectors, making it ideal for document comparison.
- Discuss scalability challenges when calculating pairwise cosine similarity for many documents. How could Spark help address these challenges?**

The main scalability challenge is $O(n^2)$ pairwise comparisons. Spark addresses this via distributed computation, memory management, and approximate similarity algorithms.

Q12

```

+-----+-----+-----+
|book1|book2|cosine_similarity|
+-----+-----+-----+
|hd fs://localhost:9000/D184MB/108.txt|hd fs://localhost:9000/D184MB/221.txt|0.9991771501221716|
|hd fs://localhost:9000/D184MB/10.txt|hd fs://localhost:9000/D184MB/30.txt|0.9987882622903789|
|hd fs://localhost:9000/D184MB/117.txt|hd fs://localhost:9000/D184MB/156.txt|0.9909735039505784|
|hd fs://localhost:9000/D184MB/133.txt|hd fs://localhost:9000/D184MB/298.txt|0.9817156238580168|
|hd fs://localhost:9000/D184MB/204.txt|hd fs://localhost:9000/D184MB/223.txt|0.9770610790733313|
|hd fs://localhost:9000/D184MB/110.txt|hd fs://localhost:9000/D184MB/133.txt|0.9755881831018351|
|hd fs://localhost:9000/D184MB/107.txt|hd fs://localhost:9000/D184MB/122.txt|0.9743994704050124|
|hd fs://localhost:9000/D184MB/227.txt|hd fs://localhost:9000/D184MB/231.txt|0.9728920452943159|
|hd fs://localhost:9000/D184MB/107.txt|hd fs://localhost:9000/D184MB/145.txt|0.9727781648892969|
|hd fs://localhost:9000/D184MB/127.txt|hd fs://localhost:9000/D184MB/129.txt|0.971423460525483|
+-----+-----+-----+
only showing top 10 rows
26/02/14 19:54:43 INFO SparkContext: SparkContext is stopping with exitCode 0 from stop at NativeMethodAccessorImpl.java:0.
26/02/14 19:54:43 INFO SparkUI: Stopped Spark web UI at http://192.168.1.240:4041

```

```

+-----+-----+
|book|total_connections|
+-----+-----+
|hd fs://localhost:9000/D184MB/219.txt|179|
|hd fs://localhost:9000/D184MB/291.txt|176|
|hd fs://localhost:9000/D184MB/244.txt|176|
|hd fs://localhost:9000/D184MB/288.txt|176|
|hd fs://localhost:9000/D184MB/243.txt|176|
+-----+-----+
only showing top 5 rows
26/02/14 19:57:56 INFO SparkContext: SparkContext is stopping with exitCode 0 from stop at NativeMethodAccessorImpl.java:0.
26/02/14 19:57:56 INFO SparkUI: Stopped Spark web UI at http://192.168.1.240:4041
26/02/14 19:57:56 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/14 19:57:56 INFO MemoryStore: MemoryStore cleared
26/02/14 19:57:56 INFO BlockManager: BlockManager stopped
26/02/14 19:57:56 INFO BlockManagerMaster: BlockManagerMaster stopped
26/02/14 19:57:56 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/02/14 19:57:56 INFO SparkContext: Successfully stopped SparkContext (Uptime: 64857 ms)
(venv) (base) admin@Aditya-2: ML RD assign 1 %

```

This creates a network:

Nodes = Books

Edges = Similarity above threshold

Highly connected nodes = influential works.

- **How Similarity Graph Was Constructed**

Cosine similarity was computed between TF-IDF vectors of books.

Edges were created between book pairs whose similarity exceeded a predefined threshold (e.g., 0.7).

This resulted in a weighted similarity network.

- **How Influence Was Measured**

Influence was approximated by:

1. Number of connections (degree centrality)
2. Higher degree $\rightarrow$ more influence
3. Could also use PageRank (advanced method)

Limitations of This Approach

1. Text Similarity $\neq$ True Influence
2. Time Ignored
3. Author Metadata Missing
4. Threshold Sensitivity